

Dimensions of flywheel Rims.

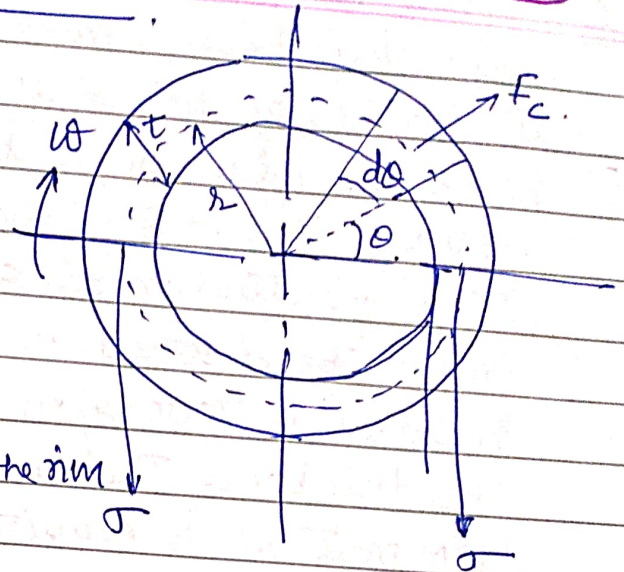
Consider a rim of flywheel.
Let

$\omega \rightarrow$ Angular Velocity

$r \rightarrow$ Mean Radius

$t \rightarrow$ Thickness of Rim

$\rho \rightarrow$ Density of the material of the rim



Consider an element of the rim,
Centrifugal force on the element/unit length $= [\rho(r.dr)t] \omega^2$

$$\begin{aligned} \text{Total vertical force/unit length} &= \int_0^\pi \rho r^2 dr \omega^2 \sin \theta = \rho r^2 t \omega^2 \int_0^\pi \sin \theta d\theta \\ &= \rho r^2 t \omega^2 (-\cos \theta)_0^\pi = 2 \rho r^2 t \omega^2 \end{aligned}$$

Let σ = Circumferential stress induced in the rim (Hoop stress)
Then for equilibrium

$$\begin{aligned} \sigma(2t).l &= 2 \rho r^2 t \omega^2 \\ \left\{ \sigma &= \rho r^2 \omega^2 = \rho v^2 \right\} \end{aligned}$$

This relation gives the limiting tangential velocity at the mean radius of the rim of the flywheel. Then diameter can be calculated by the relation.

$$v = \frac{\pi d N}{60}$$

Also, mass = Density $\times$ volume =
= Density $\times$ Circumference $\times$ Cross Sectional Area.

$$\text{or } \{ m = \rho \cdot \pi d b t \}$$

By this relation width & thickness of rim obtained.

Q1. The Turning-moment diagram for a multicylinder engine has been drawn to a vertical scale of $1 \text{ mm} = 650 \text{ N-m}$ and a horizontal scale of $1 \text{ mm} = 4.5^\circ$. The areas above and below the mean torque line are $-28, +380, -260, +310, -300, +242, -380, +265$ and -229 mm^2 .

The fluctuation of speed is limited to $\pm 1.8\%$ of the mean speed which is 400 rpm . The density of the rim material is 7000 kg/m^3 and width of the rim is 4.5 times its thickness. The Centrifugal stress (hoop stress) in the rim material is limited to 6 N/mm^2 . Neglecting the effect of the boss and arms, determine the diameter and cross section of the flywheel rim.

Q2. The torque delivered by a two stroke engine is represented by

$$T = (1000 + 300 \sin 2\theta - 500 \cos 2\theta) \text{ N-m}$$

where θ is the angle turned by the crank from the inner-dead centre. The engine speed is 250 rpm . The mass of the flywheel is 400 kg and radius of gyration 100 mm . Determine the

- Power developed
- Total percentage fluctuation of speed.
- Angular acceleration of flywheel when the crank has rotated through an angle of 60° from the inner-dead centre.
- Maximum angular acceleration and retardation of the flywheel.

Q3. A punching machine carries out 6-holes per minute. Each hole of 40-mm diameter in 35 mm thick plate requires 8 N-m of energy/ mm^2 of the sheared area. The punch has a stroke of 95 mm . Find the power

of the motor required if the mean speed of the flywheel is 20 m/s. If total fluctuation of speed is not to exceed 3% of the mean speed, determine the mass of the flywheel.

Effort of a Governor $\div \rightarrow$

The effort of the governor is the mean force acting on the sleeve to raise or lower it for a given change of speed. At constant speed, the governor is in equilibrium and the resultant force acting on the sleeve is zero. However, when the speed of the governor increases or decreases, a force is on the sleeve which tends to move it. When the sleeve occupies a new steady position, the resultant force acting on it again becomes zero.

If the force acting at the sleeve changes gradually from zero to a value E for an increased speed of the governor, the mean force or the effort is $\frac{E}{2}$.

For a Porter governor, the height is given by

$$h = \frac{g}{\omega^2} + \frac{Mg(1+k)}{2g\omega^2} = \frac{2mg + Mg(1+k)}{2m\omega^2} \quad \text{--- (A)}$$

Let ω be increased by c times ω , where c is a factor and E be the force applied on the sleeve to prevent it from moving. Thus the force on the sleeve is increased to $(Mg + E)$. Then

$$h = \frac{2mg + (Mg + E)(1+k)}{2m(1+c)^2 \omega^2} \quad \text{--- (B)}$$

Divide Eqn (A) by (B),

$$\frac{2mg + (Mg + E)(1+k)}{2mg + Mg(1+k)} = \frac{(1+c)^2}{1}$$

or
$$\frac{[2mg + (Mg + E)(1+k)] - [2mg + Mg(1+k)]}{2mg + Mg(1+k)} = \frac{1+c^2-2c-1}{1}$$

c^2 is a small quantity, neglecting it

$$\frac{E(1+k)}{2mg + Mg(1+k)} = 2c$$

or
$$E = \frac{2c}{(1+k)} [2mg + Mg(1+k)]$$

$\therefore$ Effort,

$$\left\{ \frac{E}{2} = \frac{cg}{(1+k)} [2m + M(1+k)] \right\} \text{--- (C)}$$

* If $k=1$

Effort,
$$\left\{ \frac{E}{2} = (m+M)cg \right\} \text{--- (D)}$$

* If friction of the sleeve is considered

Effort,
$$\left\{ \frac{E}{2} = (mg + Mg + f)c \right\} \text{--- (E)}$$

* For watt Governor, $M=0$

Effort
$$\left\{ \frac{E}{2} = c.m.g \right\} \text{--- (F)}$$

$\therefore$ Effort of watt governor is less than a Porter Governor

* Sometimes Effort is defined as the force required to be applied for 1% change in speed,

(3)

$$\text{Effort} = (m+M)c \cdot g = 0.01(m+M) \cdot g$$

$c = 1\%$

For a Hartnell Governor

$$mr\omega^2 a = \frac{1}{2}(Mg + F_s)b. \quad \text{--- (i)}$$

Let E be the force applied on the sleeve to prevent its movement when the speed changes from ω to $c\omega$.

$$mr(1+c)^2 \omega^2 a = \frac{1}{2}(Mg + E + F_s)b. \quad \text{--- (ii)}$$

Divide Eqⁿ (i) by (ii)

$$\frac{1}{(1+c)^2} = \frac{Mg + F_s}{Mg + E + F_s}$$

$$\text{or } \frac{Mg + E + F_s}{Mg + F_s} = (1+c)^2$$

$$\frac{E}{Mg + F_s} = 1 + c^2 + 2c - 1 = 2c \quad (\text{Neglecting } c^2).$$

$$\therefore \text{Effort} \left\{ \frac{E}{2} = c(Mg + F_s) \right\}$$

POWER OF A GOVERNOR →

The power of a governor is the work done at the sleeve for a given percentage change of speed, i.e. it is the product of the effort and the displacement of the sleeve.

For a Porter governor, having all equal arms which intersect on the axis or pivoted at

at points equidistant from the spindle axis.

$$\text{Power} = \text{Effort} \times \text{displacement of sleeve} \\ = \frac{E}{2} \times 2 \times (\text{height of governor}).$$

If the height of the governor changes from h to h_1 when the speed changes from ω to $(1+c)\omega$.

$$h = \frac{2m + Mg(1+k)}{2m\omega^2} \quad \& \quad h_1 = \frac{2m + Mg(1+k)}{2m(1+c)^2\omega^2}$$

$$\therefore \frac{h_1}{h} = \frac{1}{(1+c)^2}$$

and displacement of sleeve $= 2(h - h_1)$

$$= 2h \left(1 - \frac{h_1}{h} \right)$$

$$= 2h \left[1 - \frac{1}{(1+c)^2} \right]$$

$$= 2h \left[1 - \frac{1}{1+2c} \right] \quad \text{neglecting } c^2$$

$$= 2h \left(\frac{2c}{1+2c} \right)$$

$$\therefore \text{Power} = (m+M)cg \times 2h \cdot \left(\frac{2c}{1+2c} \right)$$

$$\left[\text{Power} = (m+M)gh \left(\frac{4c^2}{1+2c} \right) \right]$$

* In Case $k \neq 1$,

$$\Rightarrow \text{Displacement of sleeve} = (1+k)(h-h_1) = (1+k)h \left(\frac{2c}{1+2c} \right)$$

$$\Rightarrow \text{Power} = \frac{Cg}{(1+k)} \left[2m + M(1+k) \right] \times (1+k)h \left(\frac{2c}{1+2c} \right)$$

$$= \left[m + \frac{M}{2}(1+k) \right] g \cdot h \left(\frac{4c^2}{1+2c} \right)$$

Q1. Each ball of a porter governor has a mass of 3 kg and the mass of the sleeve is 15 kg. The governor has equal arms, each of 200 mm length and pivoted on the axis of rotation. When the radius of ~~gyration~~ rotation of the balls is 120 mm, the sleeve begins to rise up 160 mm at the Max^m speed. Determine

- (a) Range of speed (b) Lift of the sleeve
(c) Effort of the governor
(d) Power of the governor

What will be effect of friction at the sleeve if it is equivalent to 8 N?

Q2. In a Hartnell governor the radius of rotation of the balls is 60 mm at the minimum speed of 240 rpm. The length of the ball arm is 130 mm and the sleeve arm is 80 mm. The mass of each ball is 3 kg and the sleeve is 4 kg. The stiffness of the spring is 20 N/mm. Determine

- (i) Speed when the sleeve is lifted by 50 mm.
(ii) Initial compression of spring (iii) Governor effort
(iv) Power

- Q3 In a proell governor the mass of each ball is 8 kg and the mass of the sleeve is 120 kg. Each arm is 180 mm long. The length of extension of lower arms to which the balls are attached is 80 mm. The distance of pivots of arms from axis of rotation is 30 mm and the radius of rotation of the balls is 160 mm when the arms are inclined at 40° to the axis of rotation. Determine
- Equilibrium speed
 - Coefficient of Insensitiveness if the friction of mechanism is equivalent to 30 N.
 - Range of speed when governor is inoperative.
- Q4 Each arm of a porter governor is 250 mm long. The upper and lower arms are pivoted to links of 40 mm and 50 mm, respectively from the axis of rotation. Each ball has a mass of 5 kg and the sleeve mass is 50 kg. The force of friction on the sleeve of the mechanism is 40 N. Determine the range of speed of the governor for extreme radii of rotation of 125 mm and 150 mm.
- Q5 In a porter governor, each of the four arms is 400 mm long. The upper arms are pivoted on the axis of the sleeve whereas the lower arms are attached to the sleeve at a distance of 45 mm from the axis of rotation. Each ball has a mass of 8 kg and the load on the sleeve is 60 kg. What will be the equilibrium speeds for the two extreme radii of 250 mm and 300 mm of rotation of governor balls?

The mass of each ball of a proell governor is 7.5 kg and the load on the sleeve is 80 kg . Each of the arms is 300 mm long. The upper arms are pivotted on the axis of rotation whereas the lower arms are pivotted to links of 40 mm from the axis of rotation. The extensions of the lower arms to which the balls are attached are 100 mm and are parallel to the governor axis at the minimum radius. Determine the equilibrium speeds corresponding to extreme radii of 180 mm and 240 mm .

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